DESCRIPTIVE STATISTICS · STUDY NOTES

Measures of Variability: Variance

How spread out is your data, and how do we put a number on it?

At a glance
Variance is the average squared distance of data points from their mean. A **small** variance means values cluster tightly around the mean; a **large** one means they are widely spread. Use $\sigma^2 = \Sigma(x_i - \mu)^2 / N$ for a population and $s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$ for a sample.

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1. Where variability fits

Descriptive statistics summarise a dataset using three families of measures. Knowing the centre of the data is never enough: two datasets can share the same mean and still look completely different. Variability tells you how much the values differ from one another.

Family	Question it answers	Typical measures
1. Central tendency	Where is the centre?	Mean, median, mode
2. Asymmetry	Is the data lopsided?	Skewness
3. **Variability**	**How spread out is it?**	**Variance**, standard deviation, coefficient of variation

There are many ways to quantify variability. The three most common are **variance**, **standard deviation** and the **coefficient of variation**. These notes cover variance; the other two build directly on it.

2. Population vs sample

Before choosing a formula you must decide which kind of data you hold.

	Population	Sample
Meaning	Every member of the group you care about	A subset drawn from that group
Size	N	n
Mean	μ (mu)	$\bar{x}$ (x-bar)
Variance	σ^2 (sigma squared)	s^2
Nature of result	An exact **parameter**. You are 100% sure of it.	A **statistic**, only an approximation of the parameter

- The formulas for a sample and a population are **different**, so decide which you have first.
- If you draw **10 different samples** from the same population you get **10 different measures**. None of them is exactly the population value.

- **Example.** All of the United States is a population. California, Oregon or Michigan are each one sample from it. A statistic computed on each state will differ from state to state.

3. The mean: the reference point

Variance measures spread *around the mean*, so we always compute the mean first. The sample and population formulas look different, but they are computed in exactly the same way: add every value and divide by how many there are.

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

Population mean (N = size of population)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Sample mean (n = size of sample)

4. Variance

Definition. Variance measures the dispersion of a set of data points around their mean value.

Population variance is the sum of the squared differences between each observation and the population mean, divided by the total number of observations N.

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

Sample variance is the sum of the squared differences between each observation and the sample mean, divided by the number of observations **minus one**.

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

How to compute it, in four steps

1. Find the mean.

2. Subtract the mean from every observation (the *deviation*).

3. Square each deviation.

4. Add the squares, then divide by N (population) or $n - 1$ (sample).

> **Why do we square the differences?**
> Dispersion is a distance, and a distance cannot be negative. Squaring makes every term non-negative. Without it, deviations above the mean would cancel deviations below it and the total would always be zero. For the data 1, 2, 3, 4, 5 the raw deviations are −2, −1, 0, 1, 2, which add to 0, while the squared deviations add to 10.

5. Worked example

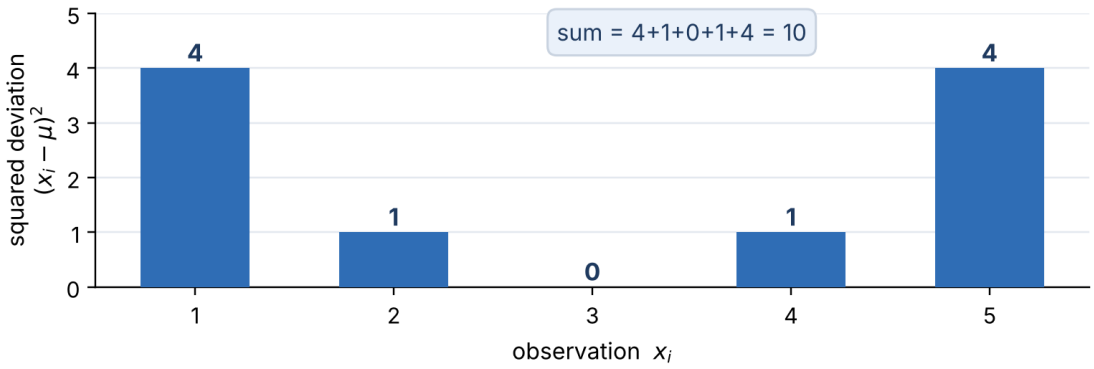
Take the five observations **1, 2, 3, 4, 5**.

Step 1: the mean

$\mu = (1 + 2 + 3 + 4 + 5) / 5 = 15 / 5 = \mathbf{3}$

Step 2: deviations and squared deviations

Observation x_i	Deviation $x_i - \mu$	Squared deviation $(x_i - \mu)^2$
1	−2	4
2	−1	1
3	0	0
4	1	1
5	2	4
Total	**0**	**10**



Step 3a: population variance (divide by N = 5)

$$\sigma^2 = \frac{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}{5} = \frac{10}{5} = 2$$

Step 3b: sample variance (divide by n − 1 = 4)

Suppose the same five numbers were instead a *sample* drawn from a bigger population. Everything is identical except the denominator.

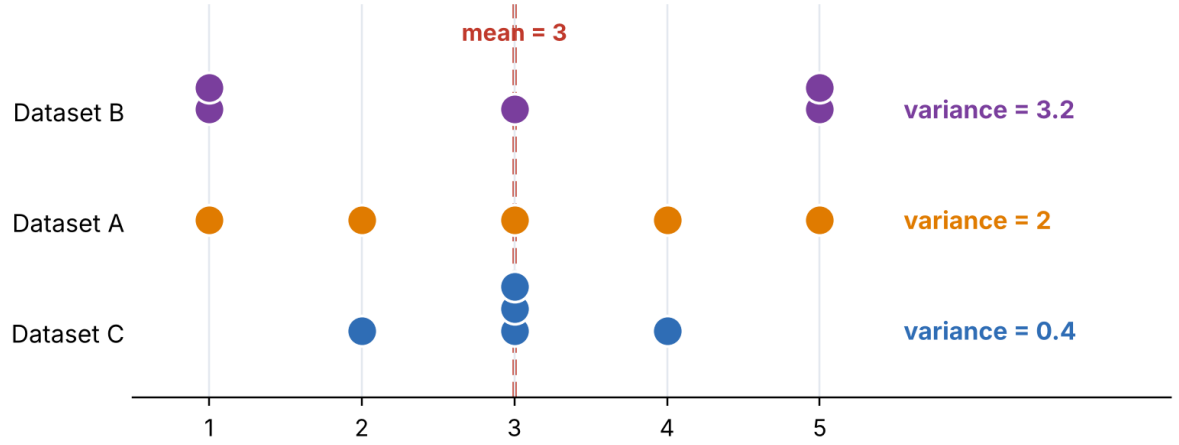
$$s^2 = \frac{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}{4} = \frac{10}{4} = 2.5$$

	Numerator	Denominator	Result
Population variance σ²	10	N = 5	**2.00**
Sample variance s²	10	n − 1 = 4	**2.50**

Note. The numbers themselves do not decide which formula to use. What matters is how the data was collected: is it the whole population, or a sample from it?

6. Reading variance: comparing spread

Variance becomes meaningful when you compare datasets. Below are three datasets that all have the **same mean (3)** but very different spread. Variances here are population variances (÷ N = 5).



Dataset	Values	Mean	Σ(x − μ)²	Variance	Interpretation
C	2, 3, 3, 3, 4	3	1+0+0+0+1 = 2	**0.4**	Tightly clustered
A	1, 2, 3, 4, 5	3	4+1+0+1+4 = 10	**2.0**	Moderate spread
B	1, 1, 3, 5, 5	3	4+4+0+4+4 = 16	**3.2**	Widely spread

The mean alone cannot tell these datasets apart. Variance can. The further values sit from the mean, the larger the variance. If every value were identical, the variance would be exactly 0.

7. Why divide by n − 1 for a sample?

The class rule is simple: **use *n − 1* whenever your data is a sample drawn from a larger population**, and use *N* only when you hold the entire population.

> **Supplementary explanation (not part of the original class notes)**
> A sample usually sits closer to its own mean $\bar{x}$ than to the true population mean μ, so dividing by *n* would systematically **underestimate** the true spread. Dividing by *n − 1* (Bessel's correction) compensates, making s^2 an unbiased estimate of σ^2. The correction matters a lot for small samples (4 instead of 5 changes 2.0 into 2.5) and shrinks as the sample grows.

8. Calculating variance in Excel and Python

Excel

Goal	Formula (data in B4:B8)	Result for 1–5
Mean	=AVERAGE(B4:B8)	3
Population variance	=VAR.P(B4:B8)	2
Sample variance	=VAR.S(B4:B8)	2.5

Python

```
import statistics as st
import numpy as np

data = [1, 2, 3, 4, 5]

st.pvariance(data)       # population variance -> 2
st.variance(data)        # sample variance     -> 2.5

np.var(data)             # ddof=0 (default) -> 2.0  (population)
np.var(data, ddof=1)     # ddof=1           -> 2.5  (sample)
```

Common pitfall
NumPy's `np.var` divides by N by default, while pandas' `Series.var()` divides by $n - 1$ by default. Always check which one you need.

9. Practice problems

Q1. The data 2, 4, 6, 8 is the *entire population*. Find the population variance.

Q2. Treat the same four numbers as a *sample*. Find the sample variance.

Q3. A dataset has variance 0. What can you say about its values?

Answers

Q	Working	Answer
1	Mean = 20/4 = 5. Squared deviations: 9, 1, 1, 9, which sum to 20. Divide by N = 4.	**5**
2	Same numerator 20. Divide by $n - 1 = 3$.	**≈ 6.67**
3	Every squared deviation must be 0, so every value equals the mean.	**All values identical**

10. Summary

- Variability describes **how spread out** data is; the mean alone cannot show this.
- **Variance** is the average squared distance of the data points from their mean.
- Population variance divides by **N**; sample variance divides by **n − 1**.
- Differences are **squared** so they are non-negative and cannot cancel out.
- A sample statistic only **approximates** the population parameter; different samples give different values.
- Variance is in **squared units** (for example dollars²), which is hard to interpret. That is why the next topic, **standard deviation**, takes the square root.

Source: class notes (Parts 2 and 3) and the accompanying Excel file *02_Variance.xlsx*. Section 6, the practice problems and the $n - 1$ explanation were added for clarity; all numbers were checked by computation.